

Marking Rubric:

Equation: 4

State Table: 9

State Diagram: 2

$$JA = B'CX + A'B$$

$$KA = A' \text{ xor } B' \text{ xor } X$$

$$DB = ABX + B'C'$$

$$TC = X'(A \text{ xnor } C')$$

State table

A	B	C	X	JA	KA	DB	TC	A+	B+	C+
0	0	0	0	0	0	1	0	0	1	0
0	0	0	1	0	1	1	0	0	1	0
0	0	1	0	0	0	0	1	0	0	0
0	0	1	1	1	1	0	0	1	0	1
0	1	0	0	1	1	0	0	1	0	0
0	1	0	1	1	0	0	0	1	0	0
0	1	1	0	1	1	0	1	1	0	0
0	1	1	1	1	0	0	0	1	0	1
1	0	0	0	0	1	1	1	0	1	1
1	0	0	1	0	0	1	0	1	1	0
1	0	1	0	0	1	0	0	0	0	1
1	0	1	1	1	0	0	0	1	0	1
1	1	0	0	0	0	0	1	1	0	1
1	1	0	1	0	1	1	0	0	1	0
1	1	1	0	0	0	0	0	1	0	1
1	1	1	1	0	1	1	0	0	1	1

STATE DIAGRAM

